Year 13 Further Maths - Differential equations in mechanics

Similar questions - Forces that depend on t and v

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle P of mass 3 kg moves along the x-axis under a constant driving force of 12 N and a resistance of magnitude $0.6v$ N, where v m s^{-1} is the velocity at time t seconds. Initially $v = 5$ and $x = 0$.

(a) Show that $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 4 - 0.2v$. **[2]**

(b) Solve to show that $v = 20 - 15\mathrm{e}^{-0.2t}$. **[5]**

(c) Find the exact displacement after 10 seconds. **[3]**

(d) Describe the motion if the initial speed were 20, 12 or 30 m s^{-1}. **[3]**

2 A particle of mass 0.5 kg starts from rest at the origin on the x-axis. Two forces act along the axis: F_1 in the direction of motion, proportional to t with constant 2, and a resistive force F_2 equal to v (with v m s^{-1} the speed at time t seconds).

(a) Show that $\dfrac{\mathrm{d}v}{\mathrm{d}t} + 2v = 4t$. **[2]**

(b) Find v in terms of t. **[6]**

(c) Find an approximation for v when t is large. **[1]**

(d) In a refined model F_1 is a constant 2 N. Write down the differential equation and state what happens to v in the long term. **[2]**

3 A particle P of mass 3 kg moves on the x-axis under the action of a force acting in the positive x-direction, of magnitude $\dfrac{12}{(t+1)^2}$ N, where $t \geqslant 0$. When $t = 0$, $x = 0$ and $v = 1$.

(a) Show that $v = 5 - \dfrac{4}{t+1}$. **[4]**

(b) Find x in terms of t. **[3]**

(c) As t increases, v approaches a limiting value v_T. Find v_T and the distance P has travelled when $v = 0.95v_T$. **[5]**

4 A particle of mass 2 kg is moving on a straight line under the action of a force of magnitude $\dfrac{6}{1+\mathrm{e}^{-t}}$ N in the direction of motion, for $t \geqslant 0$. When $t = 0$ its speed is 1.9 m s^{-1}.

(a) Find the exact impulse of the force over $0 \leqslant t \leqslant 2$. **[4]**

(b) Find the speed at $t = 2$. **[2]**

(c) Find the work done by the force over this period. **[3]**

Question 1

(a) $3\dfrac{\mathrm{d}v}{\mathrm{d}t} = 12 - 0.6v$, so $\dfrac{\mathrm{d}v}{\mathrm{d}t} = 4 - 0.2v$

(b) $\displaystyle\int \frac{\mathrm{d}v}{20-v} = \int 0.2\,\mathrm{d}t$ gives $-\ln|20-v| = 0.2t + c$; $v = 20 - A\mathrm{e}^{-0.2t}$, $A = 15$

(c) $x = \int_0^{10} \left(20 - 15\mathrm{e}^{-0.2t}\right) \mathrm{d}t = 200 + 75\left(\mathrm{e}^{-2} - 1\right) = 135$ m (3 s.f.)

(d) $u = 20$: constant speed 20 (terminal velocity). $u = 12$: speed increases towards 20. $u = 30$: speed decreases towards 20

Question 2

(a) $0.5\dfrac{\mathrm{d}v}{\mathrm{d}t} = 2t - v$, so $\dfrac{\mathrm{d}v}{\mathrm{d}t} + 2v = 4t$

(b) IF e^{2t}: $\left(\mathrm{e}^{2t}v\right)' = 4t\mathrm{e}^{2t}$, $\mathrm{e}^{2t}v = 2t\mathrm{e}^{2t} - \mathrm{e}^{2t} + c$; $v(0) = 0 \Rightarrow c = 1$. $v = 2t - 1 + \mathrm{e}^{-2t}$

(c) $v \approx 2t - 1$

(d) $\dfrac{\mathrm{d}v}{\mathrm{d}t} + 2v = 4$; $v \to 2\ \mathrm{m\,s^{-1}}$ in the long term

Question 3

(a) $3\dfrac{\mathrm{d}v}{\mathrm{d}t} = \dfrac{12}{(t+1)^2}$, so $v = -\dfrac{4}{t+1} + c$; $v(0) = 1 \Rightarrow c = 5$

(b) $x = \displaystyle\int \left(5 - \frac{4}{t+1}\right) \mathrm{d}t = 5t - 4\ln(t+1)$ (since $x(0) = 0$)

(c) $v_T = 5$. $v = 4.75 \Rightarrow \dfrac{4}{t+1} = 0.25$, $t = 15$; $x = 75 - 4\ln 16 = 63.9$ m

Question 4

(a) $I = \displaystyle\int_0^2 \frac{6}{1+\mathrm{e}^{-t}}\mathrm{d}t = \int_0^2 \frac{6\mathrm{e}^t}{\mathrm{e}^t+1}\mathrm{d}t = 6\left[\ln(\mathrm{e}^t+1)\right]_0^2 = 6\ln\dfrac{\mathrm{e}^2+1}{2}$ N s $(= 8.60)$

(b) $2v - 2(1.9) = 8.60$, $v = 6.20\ \mathrm{m\,s^{-1}}$

(c) Work $= \frac{1}{2}(2)\left(6.20^2 - 1.9^2\right) = 34.8$ J